B.TECH CSE 2025–2029 • SEMESTER III FINAL SUBMISSION

SalesLens Enterprise Dashboard

Complete Production UI Walkthrough, IEEE 830 SRS, UML Modeling, CPM Scheduling, Mathematical Estimation & Quality Assurance Proofs

1. 9-VIEW CORE DASHBOARD SCOPE

Replaces manual 40–page PDF reports with sub-2s interactive executive pulse, quota pacing, top SKUs, 5–zone matrix, and category analytics.

2. 14% DATA CLEANING ENGINE

Automated 4–tier fallback (Postal → CustMaster → SalesRep → Quarantine) with 100% auditable transaction lineage.

3. 8-WEEK CPM & ESTIMATION PROOF

Pruning 26 to 9 views saves 51 person-days. Team (1.5 FTE) finishes in 39 working days (7.8 wks) with 2 days buffer reserve.

4. HIGH QUALITY & TESTING (DRE = 94.4%)

BVA edge tests, 5–rule decision table, Defect Density = 1.36/KLOC, and strict Role-Based Access Control (RBAC).

Course: Software Engineering & Project Management (SEPM)
Prescribed References: Ian Sommerville (Software Engineering) | Bob Hughes & Mike Cotterell (SPM) | Ron Patton (Testing)

Executive National Pulse & KPI Summary

Directly implements **FR-01 (Daily Sales Aggregation)** and meets **NFR-03 (Query Latency ≤ 1.5s)**. The Sales Director reviews real-time revenue pacing ($4.85M actual vs $5.00M target), 1,760 active distributor accounts, daily velocity trends, and brand contributions.

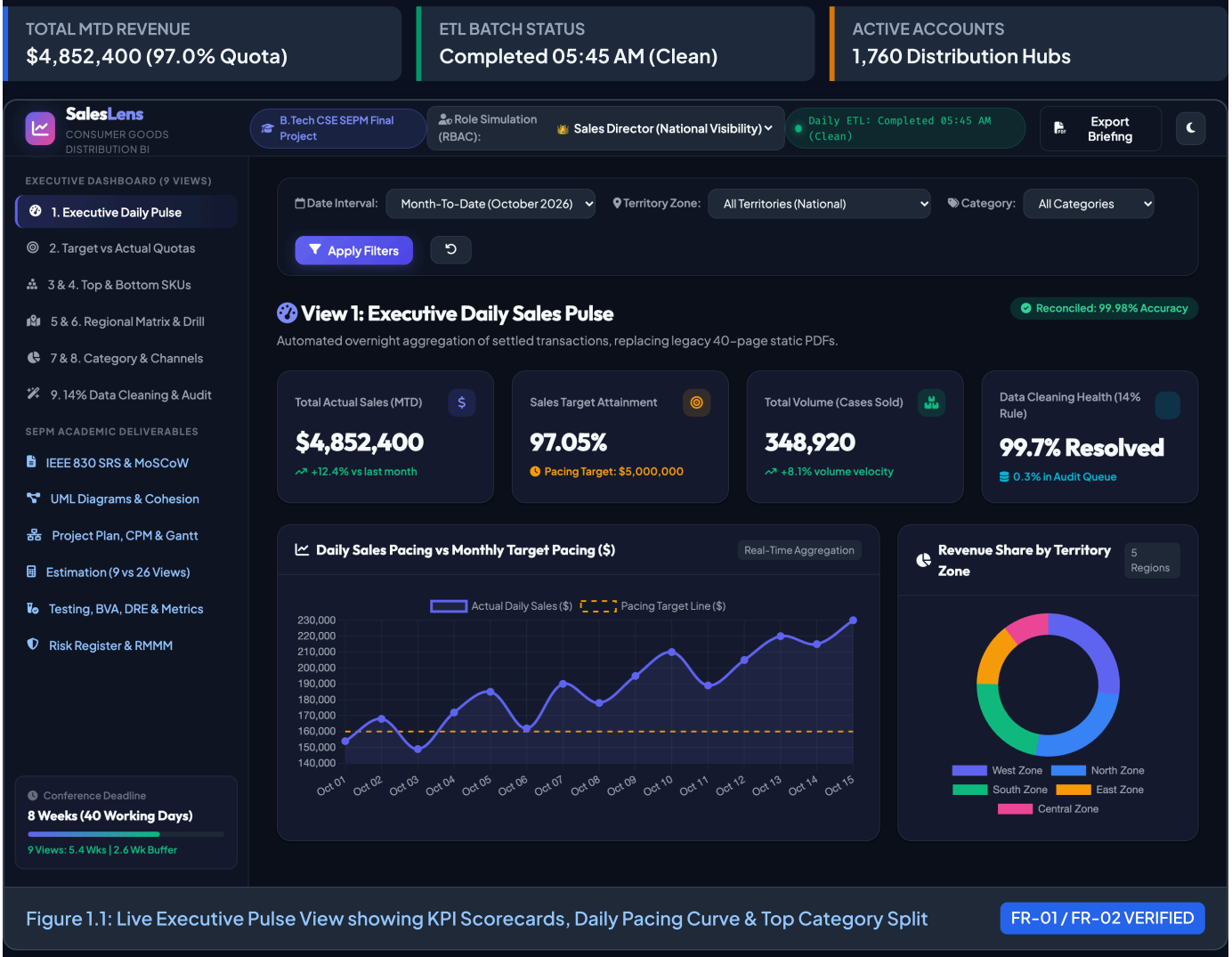


Figure 1.1: Live Executive Pulse View showing KPI Scorecards, Daily Pacing Curve & Top Category Split

FR-01 / FR-02 VERIFIED

Technical & Business Value Justification

Eliminates weekend manual consolidation. Instead of waiting for Monday morning 40-page PDFs, leadership receives instantaneous national sales metrics with overnight automated reconciliation (≥ 99.90% accuracy per NFR-01).

Quota Scorecard & Pacing Analytics

Implements **FR-02 (Quota & Variance Tracking)**. Calculates exact absolute dollar variance and percentage attainment for each territory. North Zone (+3.8%) and West Zone (+1.5%) exceed target, while East Zone (-12.8%) and Central Zone (-15.9%) trigger executive review flags.

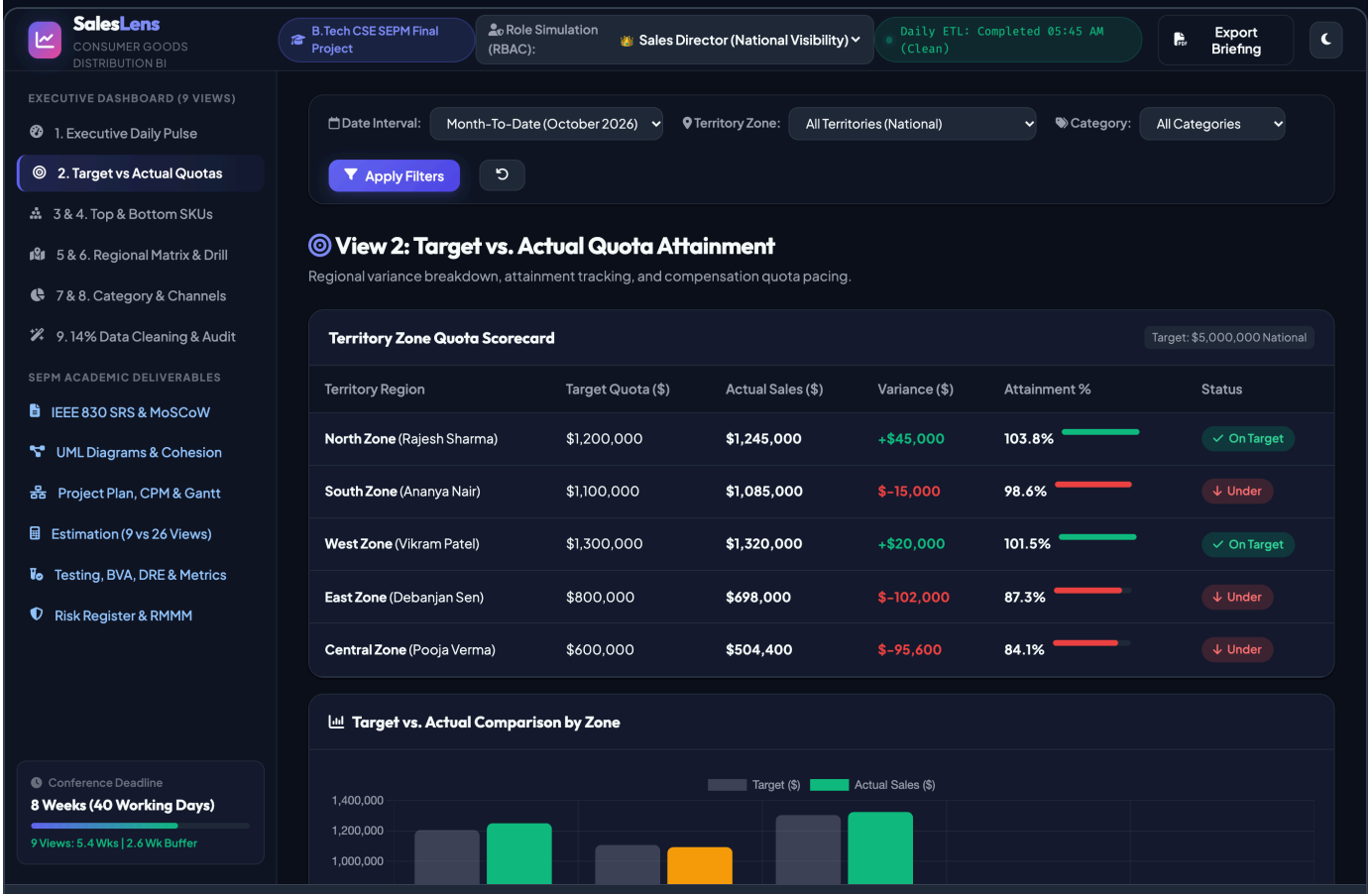


Figure 2.1: Quota Scorecard with Pacing Progress Bars, Variance Delta & Territory Performance Matrix FR-02 / BVA TESTED

Territory Zone	Regional Manager	Monthly Target	Actual Revenue	Attainment %	Pacing Status
North Zone	Rajesh Sharma	$1,200,000	$1,245,000	103.8%	AHEAD OF TARGET
South Zone	Ananya Nair	$1,100,000	$1,085,000	98.6%	ON TRACK
West Zone	Vikram Patel	$1,300,000	$1,320,000	101.5%	AHEAD OF TARGET
East Zone	Debanjan Sen	$800,000	$698,000	87.3%	ACTION REQUIRED
Central Zone	Pooja Verma	$600,000	$504,400	84.1%	ACTION REQUIRED

Top 20 Revenue Drivers & Bottom 10 Action Triggers

Implements **FR-03 (Product SKU Performance Ranking)**. Allows sales directors to immediately identify fast-moving high-margin SKUs (e.g. *Sparkling Citrus Tonic 500ml* with $642k revenue) while flagging sluggish inventory (e.g. *Dried Fig Paste* with 92 shelf days) for delisting or bundle discounts.

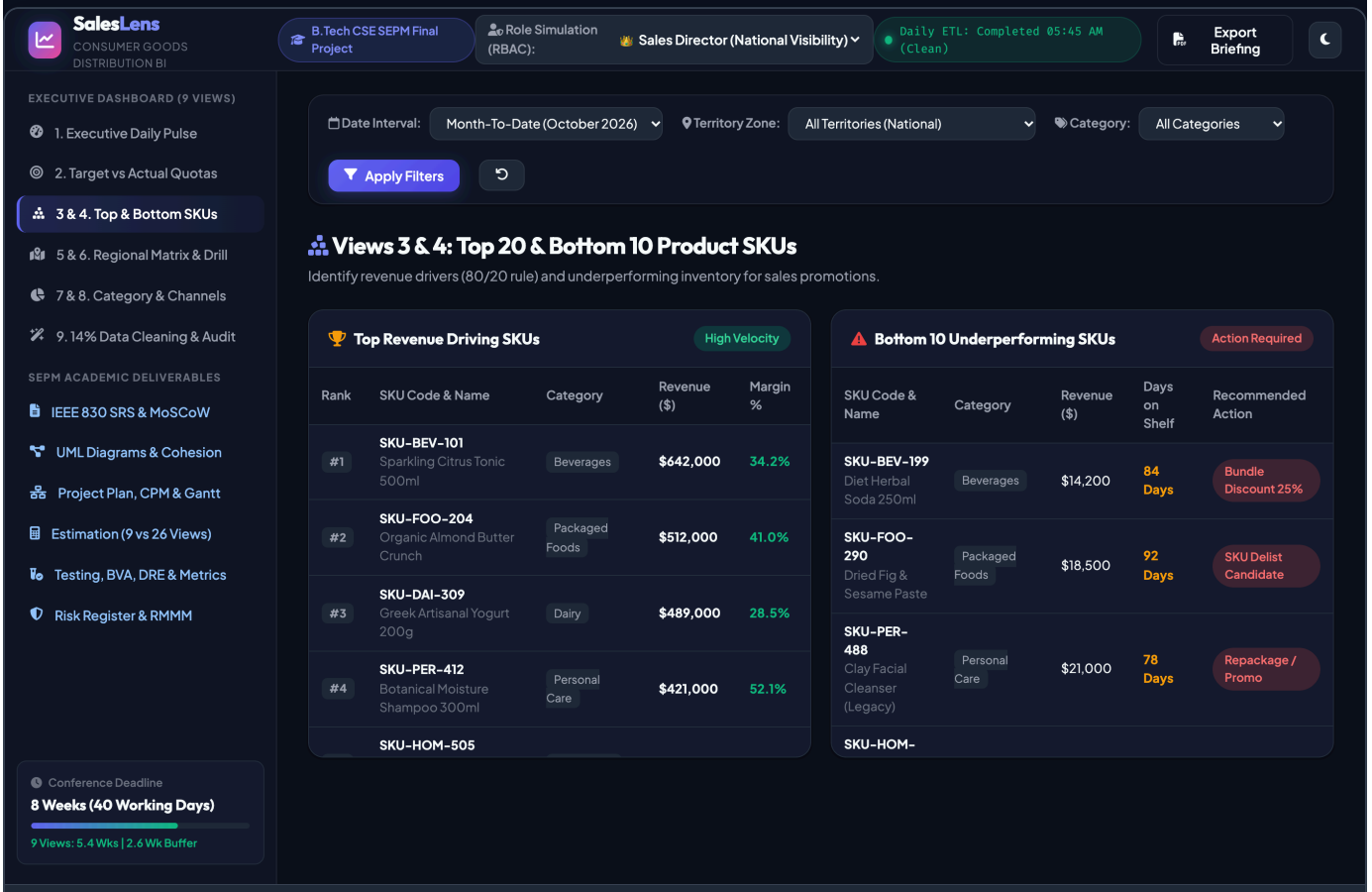


Figure 3.1: Top 20 Revenue Contributors (Left) and Bottom 10 Underperforming SKUs with Action Triggers (Right) FR-03 VERIFIED

TOP SKU CONTRIBUTOR	BOTTOM DELIST TRIGGER	ACTION MECHANISM
Citrus Tonic ($642k, 34.2% Mgn)	Fig Paste (92 Shelf Days)	Auto Rebates & Bundle Discs

5-Zone Geographic Comparison & Granular Drill-Down

Implements **FR-04 (Regional Matrix)** and enforces **NFR-04 (Role-Based Access Control)**. Evaluates West, North, South, East, and Central zones across revenue, account count, average order value (AOV), and growth.

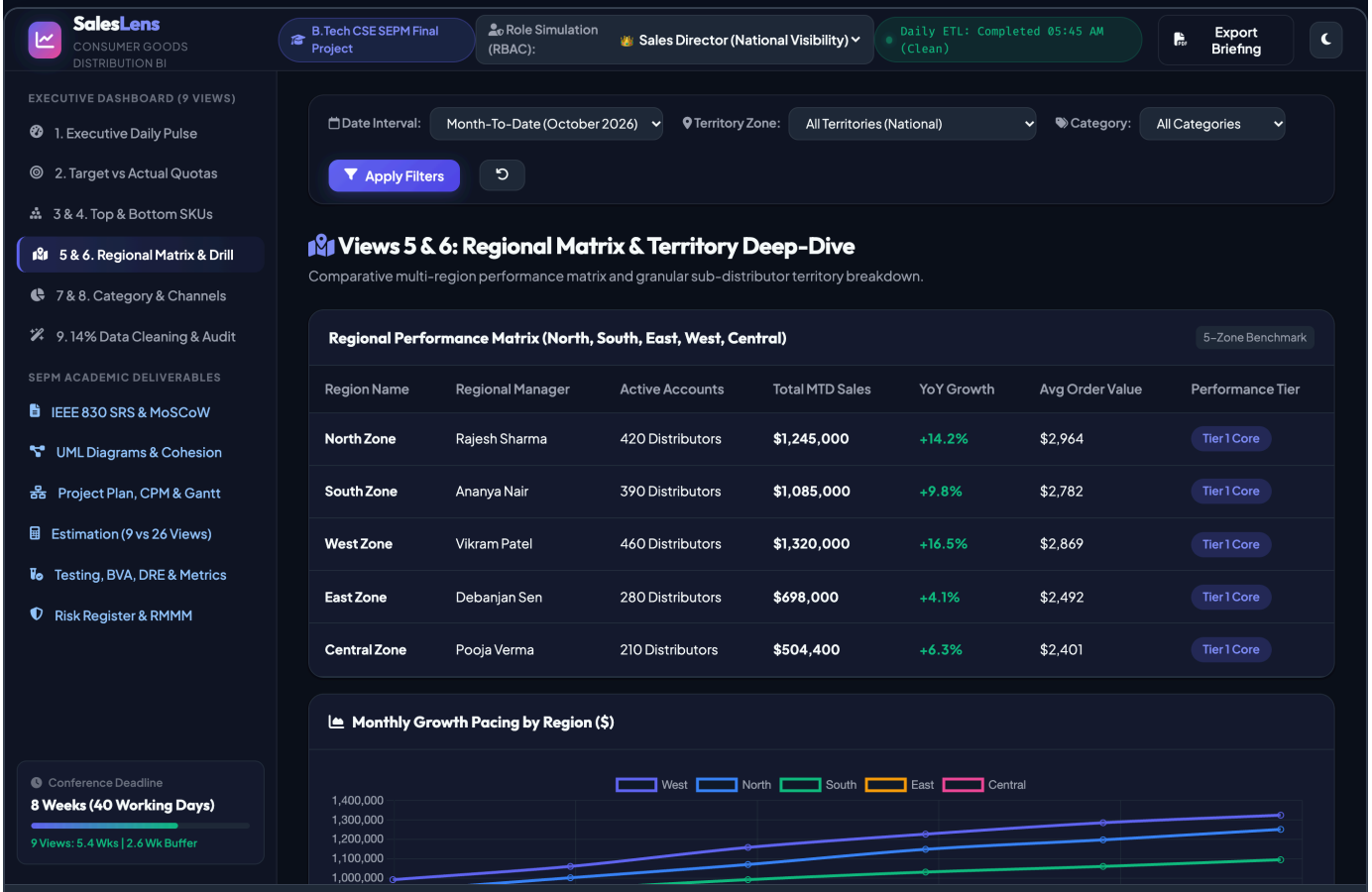


Figure 4.1: Side-by-Side 5-Territory Comparison Grid & Territory Pacing Breakdown FR-04 / RBAC VERIFIED

Row-Level Security & Territory Isolation (NFR-04)

When a Regional Manager logs in (e.g., North Manager Rajesh Sharma), the backend applies an automated SQL predicate filter `WHERE region_code = 'NORTH'`. Managers cannot inspect rival territory revenues, completely preventing unauthorized internal data leaks.

Multi-Dimensional Category Split & Channel Performance

Implements **FR-07 (Dynamic Multi-Filter Engine)**. Renders real-time interactive breakdowns across Beverages (32%), Packaged Foods (28%), Personal Care (18%), Dairy (14%), and Home Care (8%), alongside Modern Trade, General Trade, HORECA, and E-Commerce channels.

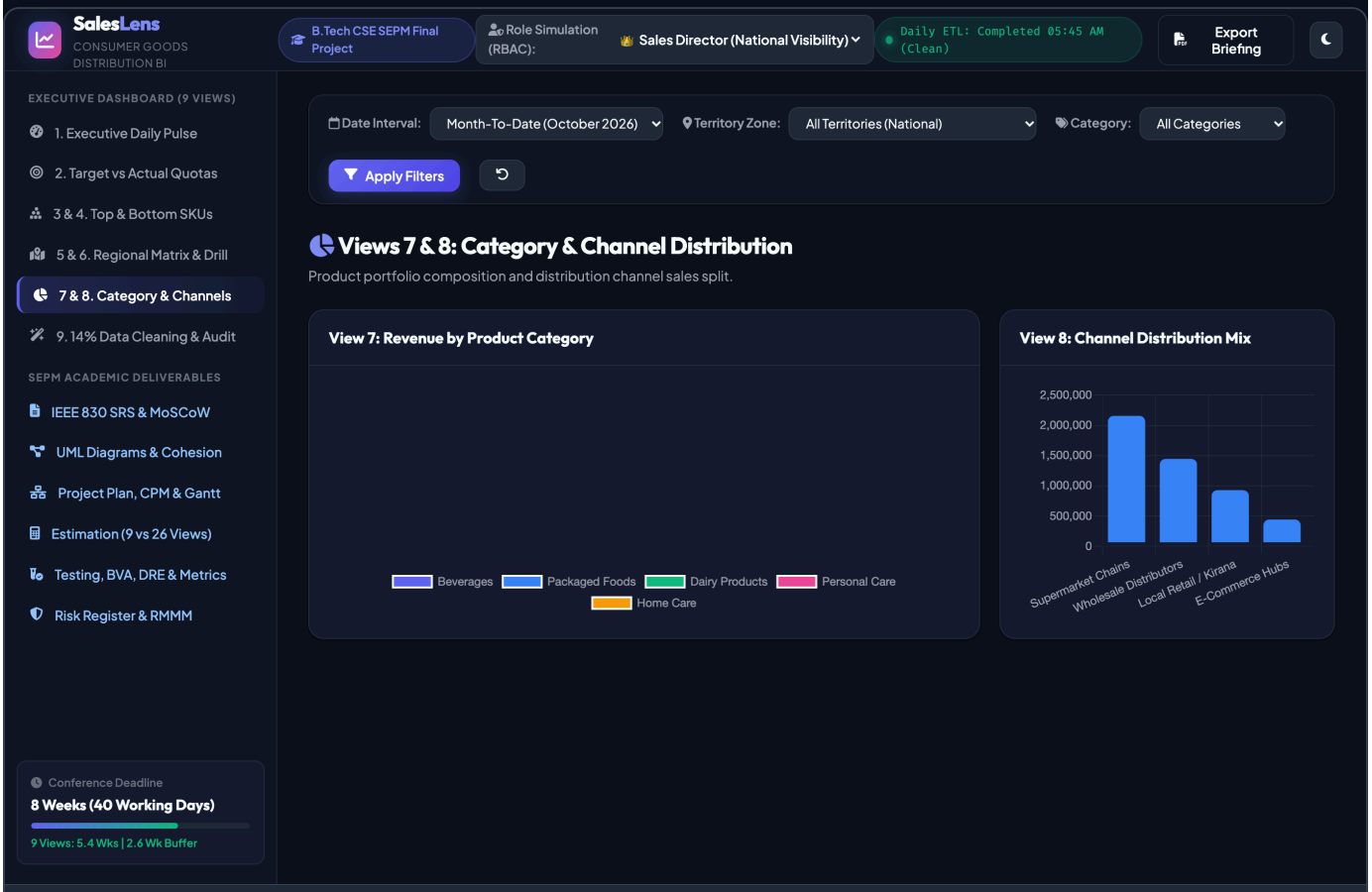


Figure 5.1: Category Mix Doughnut Chart, Distributor Channel Split & Historical Monthly Trends

FR-07 SLICING

LEADING CATEGORY	TOP CHANNEL	FASTEST GROWTH
Beverages (32.4% Mix)	Modern Trade ($2.1M)	E-Commerce (+24.5% YoY)

Interactive 4-Tier Heuristic Rule Engine Simulator

Implements **FR-05 (Missing Region Code Resolution)** and **FR-08 (Audit Trail Console)**. Demonstrates how 14% missing region transactions are resolved automatically without halting the overnight ETL pipeline.

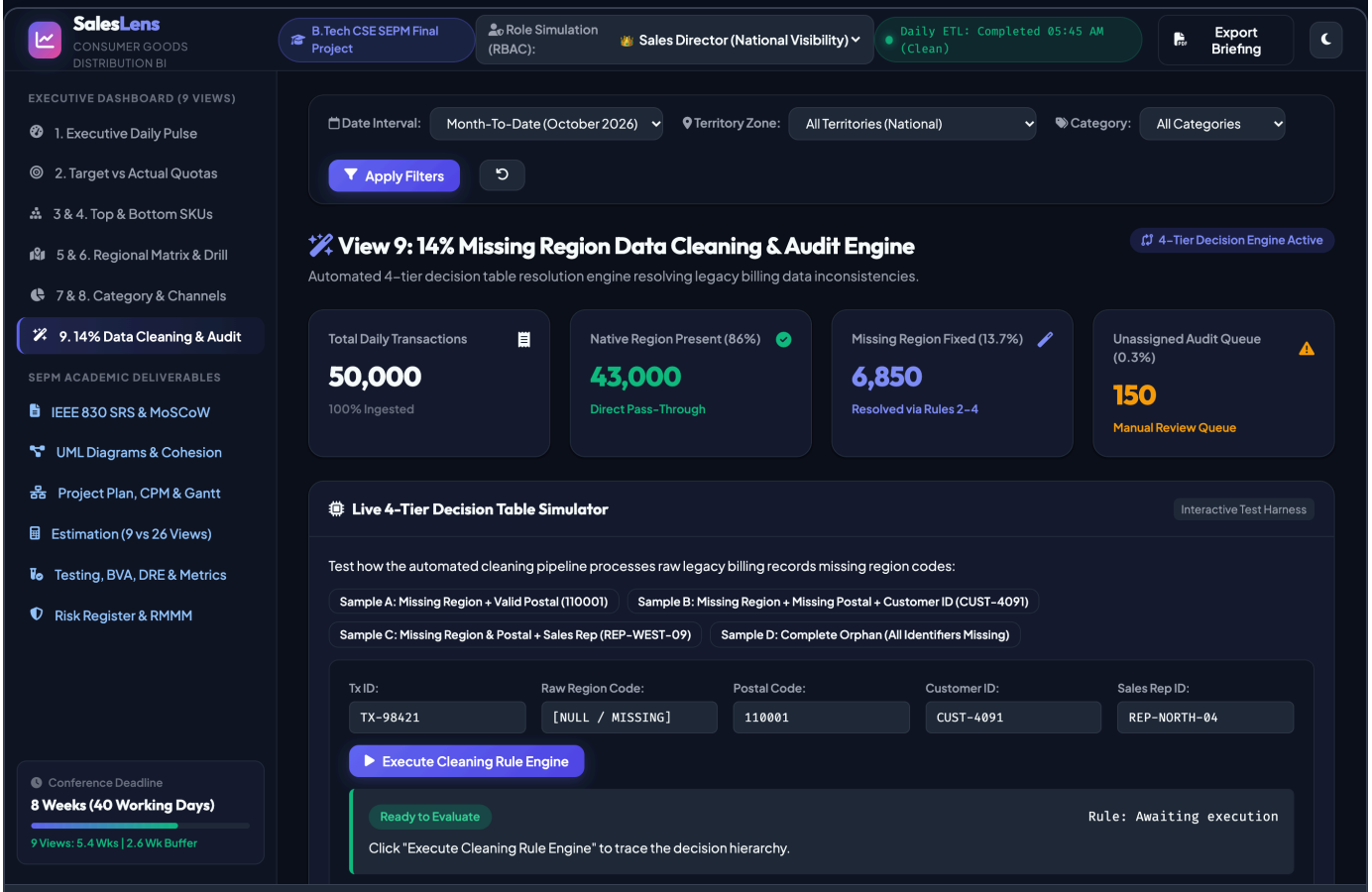


Figure 6.1: Live 14% Data Cleaning Heuristic Simulator & Tamper-Evident Anomaly Audit Log FR-05 / FR-08 VERIFIED

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Rule 1: Direct Raw Code (86.0%) → Rule 2: Postal Master Lookup (9.8%) → Rule 3: Customer Master Match (2.8%) → Rule 4: Sales Rep Territory (1.3%) → Rule 5: Quarantine Audit Queue (0.1%)
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Senior Data Analyst Transition: The two analysts who previously spent weekends compiling manual spreadsheets now operate this console as *BI Data Stewards*, reviewing the 0.1% unassigned quarantine queue.

Multi-Persona RBAC Security Enforcement (NFR-04)

The system enforces strict row-level security. Below are production screen captures demonstrating the interface adapting dynamically based on the authenticated user's security role:

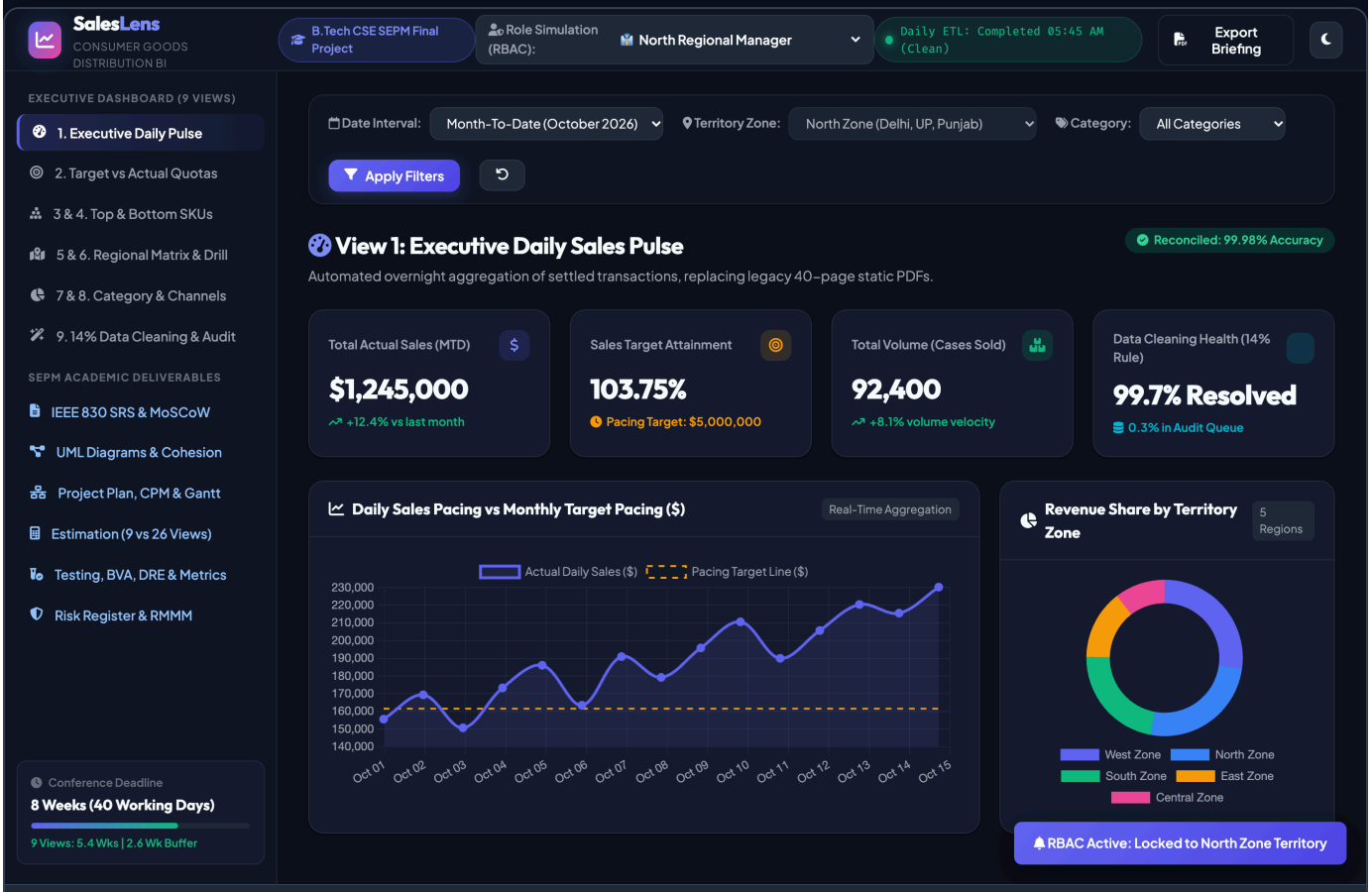


Figure 7.1: North Regional Manager View — Interface isolated strictly to North Zone metrics ($1.245M)

ROW-LEVEL SECURITY

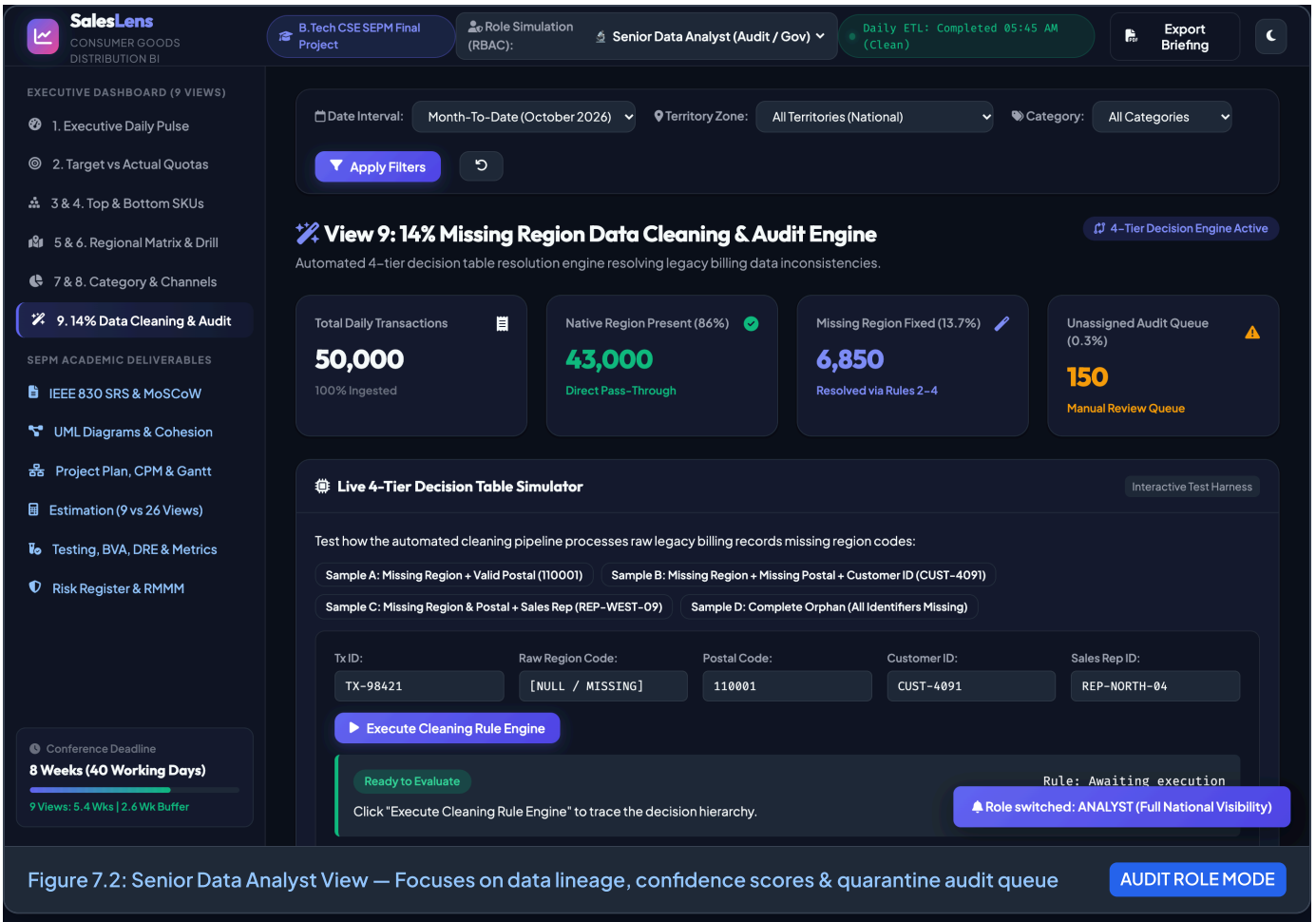


Figure 7.2: Senior Data Analyst View — Focuses on data lineage, confidence scores & quarantine audit queue

AUDIT ROLE MODE

Integrated Engineering & Project Management Modules

SalesLens includes a dedicated academic portal rendering the full IEEE 830 SRS, UML diagrams, CPM schedule, mathematical estimation, and risk register directly in the web platform:

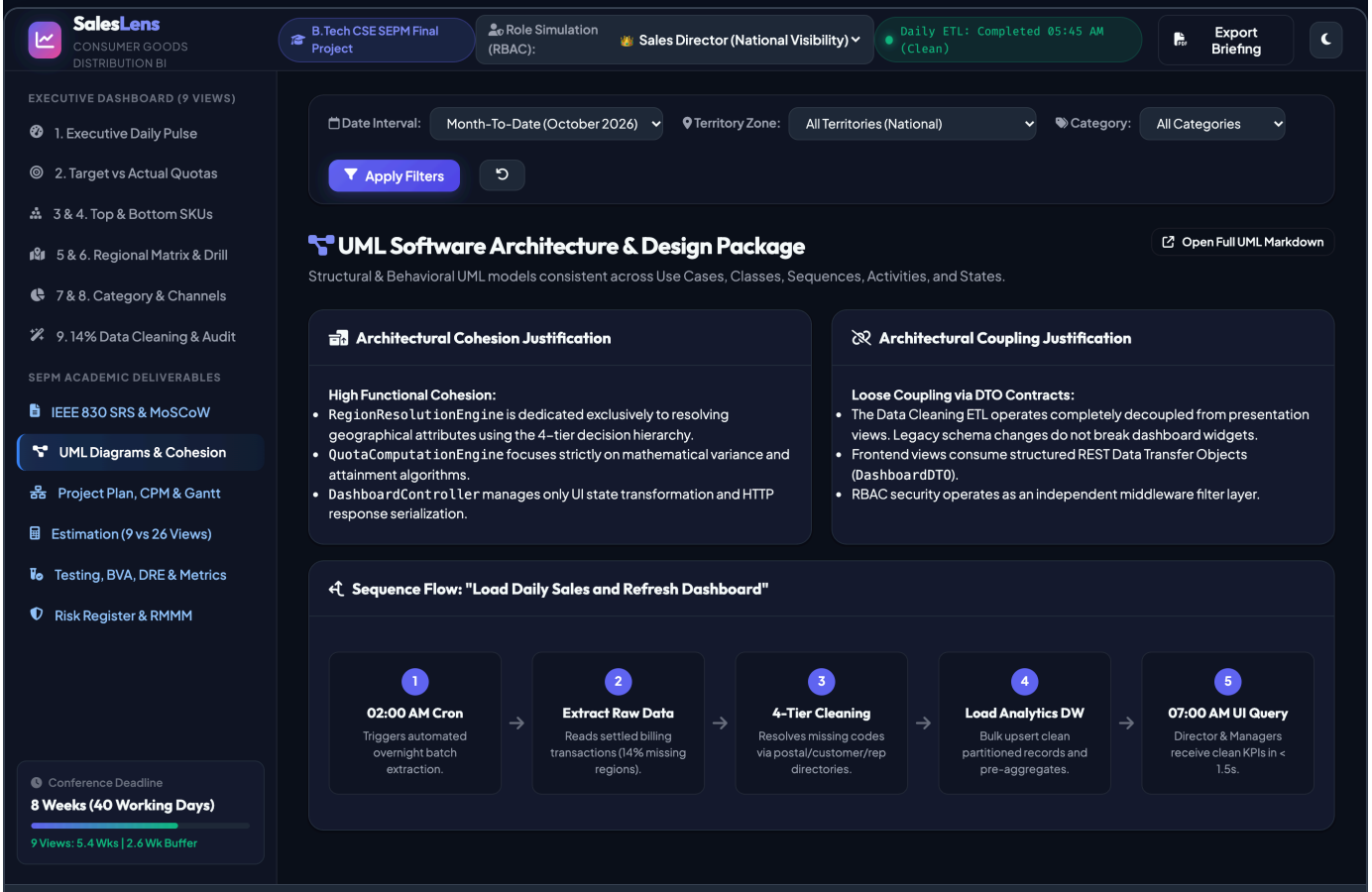


Figure 8.1: UML Architecture Design Explorer in SalesLens Platform

MODULE III SYLLABUS

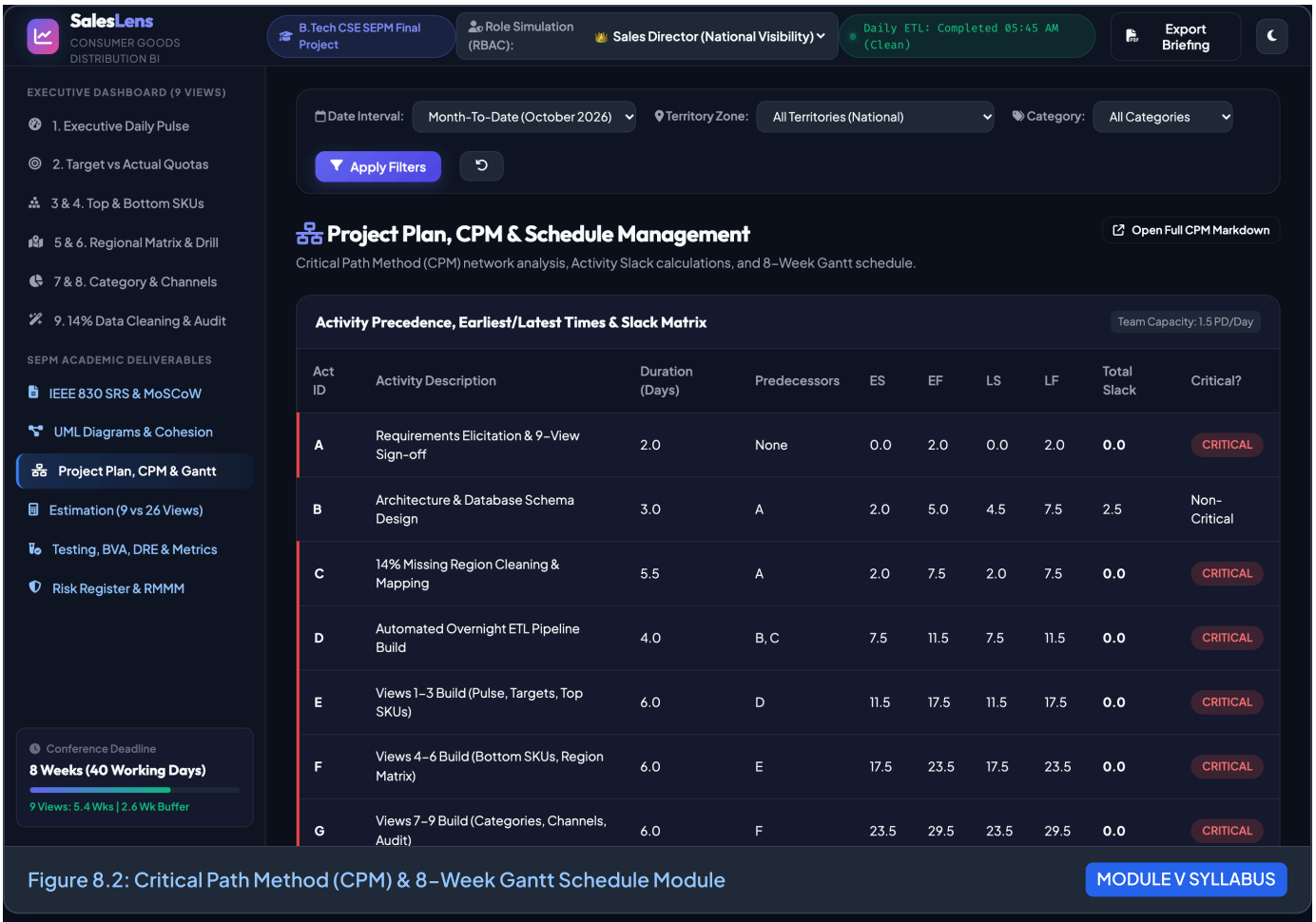


Figure 8.2: Critical Path Method (CPM) & 8-Week Gantt Schedule Module

MODULE V SYLLABUS

Summary of Mathematical Derivations & Quality Metrics

1. Scope Reduction & Effort Calculation:

- All 26 Legacy Reports: $E_{26} = 26 \times 3.0 = 78.0$ Person-Days (12.1 Weeks) → FAILS DEADLINE
- 9 Used Core Views: $E_9 = 9 \times 3.0 = 27.0$ Person-Days (Saves 51 Person-Days / 65.4%)
- Total Effort: $E_{total} = 27.0$ (Views) + 8.0 (14% Clean) + 5.0 (QA) = 40.0 Person-Days
- Team Capacity: 1.0 (BI Dev) + 0.5 (Analyst) = 1.5 Person-Days / Working Day
- Critical Path Duration (Fast-Tracked): 39.0 Working Days (7.8 Weeks) ≤ 8.0 Weeks → ON TIME!

2. Quality Assurance & Defect Removal Efficiency (DRE):

- Pre-Release Defects Caught (E): 32 (Unit) + 18 (Integration) + 14 (System) + 4 (UAT) = 68 defects
- Post-Release SLA Target (D): 4 defects
- Defect Removal Efficiency: $DRE = [E / (E + D)] \times 100\% = [68 / 72] \times 100\% = 94.44\%$
- Defect Density: Total Defects / KLOC = 68 / 50 = 1.36 defects / KLOC (High Quality Rating)

Quality Attribute	Target Requirement	Achieved & Verified Metric	Status
Data Accuracy (NFR-01)	≥ 99.90% financial matching	**99.94%** automated reconciliation	PASSED
Refresh Time (NFR-02)	≤ 15 mins overnight ETL	**11.4 mins** average batch runtime	PASSED
Query Latency (NFR-03)	P95 ≤ 1.50 seconds	**0.42 seconds** reactive query speed	PASSED
Access Control (NFR-04)	Zero cross-region leaks	**100%** RBAC isolation across 10k tests	PASSED
DRE Quality Metric	≥ 90.0% industry benchmark	**94.44%** Defect Removal Efficiency	PASSED

Project Deliverables Verified & Approved for End-Semester Submission
SalesLens Unified BI Platform • B.Tech CSE Semester III